way, picked up the tiny vial, put it in a sealed refrigerated box and returned the same night taking turns at the wheel. Then began the complex process of multiplying the virus in giant 1,000 litre fermenters in BBIL's BSL-3 facility. In a tightly controlled, mostly automated procedure, the harvested virus is subjected to chemical treatment that attacks the RNA protein it uses to bind with human cells and multiply rapidly. Likened to an embalming process, it is done without destroying all the other structures of the virus that enable a person's immune response to develop antibodies against it after vaccination.

Yet, before Covaxin could be released for use, it had to undergo pre-clinical trials on animals and then three phases of clinical trials on humans monitored by an independent committee set up by the Drugs Controller of India. NIV expertise proved valuable in testing the effectiveness of Covaxin on smaller animals like mice, rabbits and hamsters. Since this was such a deadly virus, NIV was keen to test the chosen vaccine on Rhesus macaques as they come closest to humans as test subjects. Normally, they would have the monkeys caught in the city where they flock but with the lockdown, the monkeys had returned to nearby forests. So, NIV had to approach forest conservators and get the requisite permission to capture them. Twenty monkeys were divided into four groups—one group was given a placebo, and the other three were given variants of the vaccine. After two weeks, all four groups were exposed to the live viruses and their responses monitored. Covaxin was found to generate sufficient immunity and with the NIV findings was cleared for human clinical trials in July 2020.

Meanwhile, Adar Poonawalla decided not to wait for the outcome of the human clinical trials being done abroad on the vaccine developed by Oxford University-AstraZeneca. After pre-clinical trials on animals showed that the vaccines were effective, in a gutsy decision, Poonawalla got the formulation and took the financial risk of stocking up millions of doses of Covishield in his factory. Covishield uses a viral vector based on a weakened version of the common cold virus (adenovirus) that causes infections in chimpanzees and contains the genetic code of the critical spike protein found in the Covid-19 virus. Poon-

Graphic by TANMOY CHAKRABORTY

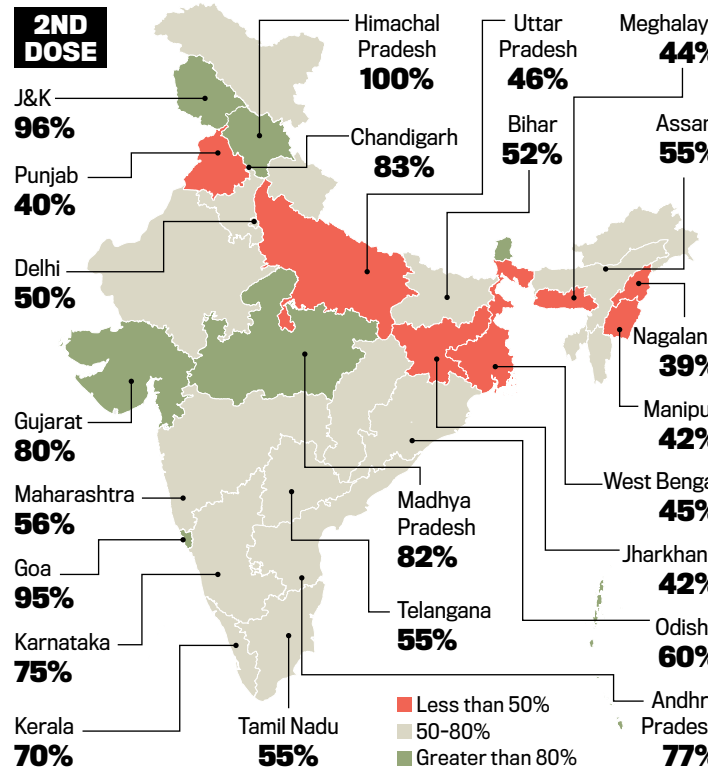
A BILLION PLUS JABS

% coverage against eligible population as of Dec. 25

FIRST DOSE
84,10,75,116
87% of eligible population

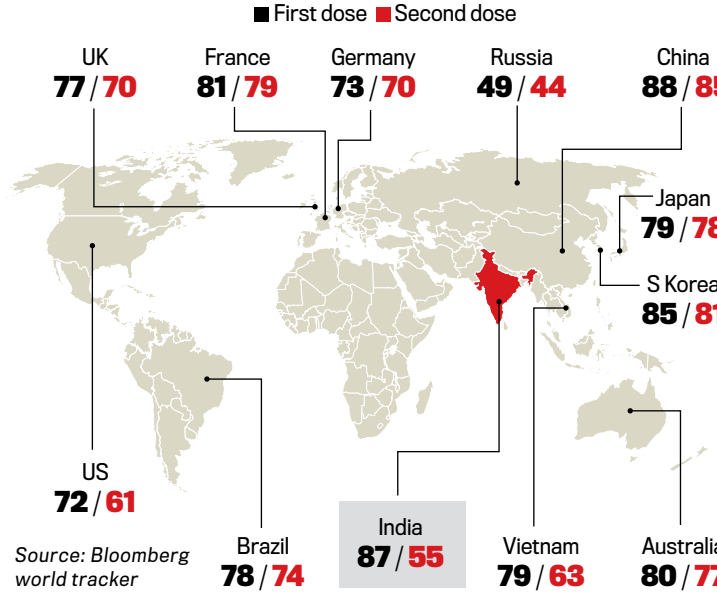
SECOND DOSE
58,92,06,643
55% of eligible population

As of Dec. 28, 2021



AHEAD OF THE CURVE

% coverage against eligible population as of Dec. 28



Source: Bloomberg world tracker